Lesson 35: The APA Research Report 2: Method, Results, Discussion and References

English Stage 4 (Psychology) · Unit 5: Writing and Presenting Research in APA Style

Adapted from Research Methods in Psychology (4th American edition) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0): 11.3 Writing a Research Report in American Psychological Association (APA) Style (from "Method" to the end of "Sample APA-Style Research Report"); exercise from 11.5 Presenting Your Research (Summary).

In this lesson

This lesson completes the research report begun in Lesson 34: a method detailed enough for replication, results organized around the research questions, a discussion, and the material at the end. In English, you practise the tense and voice of method writing, for a finished study and for a proposal, and the cautious language of limitations.

Learning goals

After this lesson you should be able to:

- Describe the subsections of a method section and choose an organization for it.
- Organize a results section with preliminary issues first and Bem's five steps for each result.
- Plan a discussion with a summary, implications, two or three limitations and suggestions for future research.
- Order the references, and use the right tense, voice and hedges in each section.

Key language

Term or phrase	Meaning	Example
method section	the section that describes the method in detail; at minimum a participants subsection and a design and procedure subsection	"Method" (not "Methods")
replication	conducting a study again, exactly or with modifications, to see whether the same results are observed	the method as a "recipe"
design	the overall structure of a study: its variables, how they were manipulated or measured, their operational definitions	a between-subjects design
procedure	how the study was carried out, often in terms of what the participants did	"read a set of instructions"
raw data	data in the form in which they were originally collected	completed questionnaires

Term or phrase	Meaning	Example
manipulation check	a measure, usually at the end, that confirms the independent variable was successfully manipulated	reported early in the results
discussion	the final major section: summary, implications, limitations, suggestions for future research	
limitation	a weakness of a study that could have influenced its results	an unreliable measure
appendix	an optional section at the end of the manuscript for important supplemental material	questionnaire items
perhaps	introduces a possibility cautiously	"Perhaps the manipulation was not very effective."
could have influenced	a past possibility (modal + *have* + past participle)	"limitations that seem like they could have influenced the results"

Rules and explanations

The ideas

Method: a recipe for replication. Include everything important (participants' demographics, recruitment, random assignment, how variables were manipulated or measured, counterbalancing) and nothing irrelevant (the study took place in Classroom 37B; the questionnaire was double-sided and completed with pencils). The section follows the introduction directly, under "Method" (not "Methods"), centered. The first subsection, headed *Participants* (left-justified and in italics, says the book), gives the number of participants, women and men, their age, other relevant demographics, and how they were recruited, including any incentives.

Three ways to organize a method section (the book's figure is a picture; this table follows the text):

After Participants	Works well when
1. Design and Procedure (one subsection)	the method is simple and fits in a few paragraphs
2. Design; Procedure (two subsections)	both are complicated and need several paragraphs each
3. Materials, then the design and procedure	there are complicated materials: questionnaires, vignettes, perceptual stimuli (the heading can be "Questionnaires", "Stimuli" and so on)

- **Design** = the overall structure: the independent and dependent variables; whether the independent variable was manipulated, between or within subjects; the operational definitions.
- **Procedure** = how the study was carried out, best told as what the participants did.
- **Materials** is also the place for reliability and validity evidence: test-retest correlations, Cronbach's α.

Results. Raw data are not included, but researchers should keep them and share them on request; many journals encourage or require open data. The order is logical:

1. Preliminary issues: excluded participants or responses, and why; how multiple responses were combined into the main variables (for recall of 20 words: number correct, percentage correct, or number correct minus number incorrect?); whether the manipulation worked (manipulation checks).
2. The research questions, one at a time: general to specific, or the main question first.

For each result, Bem (2003) suggests: (1) remind the reader of the research question; (2) give the answer in words; (3) present the relevant statistics; (4) qualify the answer if necessary; (5) summarize the result. Only step 3 needs numbers: the basic results should be clear even to a reader who skips them.

Discussion. Its usual elements:

- **Summary**: a clear answer to the research question.
- **Theoretical implications**: do the results support existing theories? If not, outline possible explanations.
- **Practical implications**: how can the results be used, and by whom?
- **Limitations**: perhaps problems of validity, a weak manipulation, unreliable measures, or participants who misunderstood the task. Skip routine issues; choose two or three that could have influenced the results, explain how, and suggest remedies.
- **Future research**: two or three important unresolved issues, with possible answers and ways to study them.

A sweeping ending can work, as in Darley and Latané (1968): "If people understand the situational forces that can make them hesitate to intervene, they may better overcome them" (p. 383). But it can sound overreaching or banal; it is often better to return to the problem in your opening and say how your research addressed it.

References, appendices, tables and figures

- References: a new page, "References" centered; alphabetical by first author; with the same first author, by the second author's last name; with all authors the same, by year; all double-spaced.
- Appendix: material that would interrupt the report (stimulus words, questionnaire items, special equipment). Each starts a new page: "Appendix" if there is one; "Appendix A", "Appendix B" in order of first mention.
- Tables and figures: each on its own page, numbered in order of first mention (Table 1, Figure 1). A table has a brief title above it, with important words capitalized; a figure has a brief caption in which only the first word of each sentence is capitalized. The book puts appendices first, then tables, then figures.
- The book's sample report (by student Tomoe Suyama) is shown only as pictures; the captions note that the method follows the introduction directly, and the discussion follows the results.

> **APA 7th edition** (the book follows older editions)

- "Method", "Results" and "Discussion" are level 1 headings: centered, **bold**, title case. "Participants", "Materials" and "Procedure" are level 2 headings: flush left, **bold**, title case, not italic.
- After the references come tables, then figures, then appendices last; or place each table and figure in the text after its first mention.
- Tables and figures: the number in bold ("**Table 1**", "**Figure 1**"), the title below it in italic title case, and any explanation in a note under the table or figure. The older caption is gone.
- "References" and "Appendix" headings are bold and centered. A one-author work comes before works by the same author with co-authors.

Language focus

1. Tense and voice. A finished study is reported in the past tense, with the participants as the subject of active verbs and the passive where the doer does not matter: "the participants gave their informed consent, read a set of instructions, completed a block of four practice trials, completed a block of 20 test trials, completed two questionnaires, and were debriefed and excused." A proposal uses the future: "Participants *will give* their informed consent … and *will be debriefed* and excused." (Note the numbers: *four* in words, *20* in numerals.)

2. Hedging limitations. The discussion uses *perhaps*, *may* and modal + *have* + past participle for possible past effects:

- "*Perhaps* the manipulation was not very effective or the measures not very reliable."
- "how they *might have affected* the results"; "a different sample or different measures *might have produced* different results"; "limitations that seem like they *could have influenced* the results".

Note for Arabic speakers

- Arabic can express a possible past event with a word meaning "perhaps" and a past verb, so learners often write "might affected" or "may affect" for a past possibility. Check every limitation: modal + *have* + past participle ("might have affected").

Practice

1. Reading questions

1. What is the main principle for writing a method section?
2. What is the difference between the design and the procedure?
3. When is a separate materials subsection useful, and which statistics can go in it?
4. What should researchers do with their raw data?
5. Which three preliminary issues open the results section?
6. In Bem's five steps, which step needs numbers? What does this show?
7. How many limitations should a discussion include, and what should the writer do with each?
8. Why can a sweeping final sentence be risky, and what is often better?

2. Book exercise and a sorting task

1. "Practice: Find a recent article in a professional journal and highlight in a different color each of the following elements in the discussion: summary, theoretical implications, practical implications, limitations, and suggestions for future research." (Use an article from your literature review.)
2. Where does each item from 11.3 belong: Participants, Materials, Design, Procedure, or nowhere? (a) the number of women and men; (b) the study took place in Classroom 37B; (c) incentives for participation; (d) Cronbach's α for each questionnaire; (e) the independent variable was manipulated within subjects; (f) participants completed four practice trials; (g) the questionnaire was double-sided; (h) written vignettes.

3. Language exercise

1. Rewrite the book's procedure sentence (Language focus 1) for a proposal, in the future tense.
2. Complete with modal + *have* + past participle: (a) A different sample ____ (produce) different results. (b) Participants who were suspicious of the researchers' intent ____ (respond) differently.
3. Put these sentences in Bem's order (invented data for practice, from your project in Lessons 27 to 32):
   - (a) In short, a nap after learning improved memory for the word pairs.
   - (b) Participants in the nap condition recalled more word pairs (M = 22.00, SD = 2.54) than participants in the wake condition (M = 17.00, SD = 3.09), $t(18) = 3.95$, $p < .001$, $d = 1.86$.
   - (c) The main question was whether a nap after learning would improve recall of the word pairs.
   - (d) However, the nap took place in a laboratory, so the result may not generalize to sleep at home.
   - (e) It did: students who napped recalled more pairs than students who stayed awake.

Speak with your tutor

- "Here is my method section: [paste]. Could another researcher run my study from it? What is missing, and what is irrelevant?"
- "Name four possible limitations of my proposal. I will choose the most important two or three and explain how I would deal with them."
- "In PSYC 110 Lecture 15 (Morality, Chapter 5), Professor Bloom's advice from bystander research is to point to one person and ask them to call the police. Is this a practical implication in the book's sense? Check my two hedged sentences about it."
- "Give me a result from my invented data, and I will present it in Bem's five steps."

Writing task

Your research project: method, planned results and discussion. Turn your drafts (measures, Lesson 12; design and procedure, Lesson 16; consent and debriefing, Lesson 09; results, Lesson 32) into three sections of your proposal on sleep and memory in university students, or your own topic (900 to 1,200 words). Keep the design behind your invented data (Lessons 27 to 32): 20 students

randomly assigned to a nap or a wake condition after learning 40 word pairs, with recall out of 40 as the dependent variable.

1. **Method** (400 to 600 words, future tense): Participants (planned number, recruitment, incentives, the demographics you will report); Materials (each measure, its operational definition, reliability or validity evidence from your sources); Design and Procedure (variables, between or within subjects, random assignment or counterbalancing, each step from informed consent to debriefing).
2. **Planned Results** (150 to 300 words): the preliminary issues (exclusions, manipulation check), the planned analysis for each hypothesis, and your Lesson 32 results as an illustration in Bem's five steps, labelled "invented data for practice".
3. **Discussion** (250 to 400 words): what the expected result would mean for theory and practice; two or three limitations, each with a remedy; one or two directions for future research; a last sentence that returns to your opening.

Self-editing checklist:

- Level 1 headings are centered and bold; level 2 headings are flush left and bold.
- Another researcher could run the study from my method, and it contains no irrelevant details.
- The procedure says what participants will do, in the future tense.
- Every measure has an operational definition and, where possible, cited reliability or validity evidence.
- Each planned result follows Bem's steps and is clear without the numbers; invented numbers are labelled.
- Statistics use APA symbols in italics (*M*, *SD*, *t*, *p*).
- Limitations are hedged (*may*, *might have*, *perhaps*) and each one has a remedy.

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Answer key

1. Reading questions

1. It should be clear and detailed enough for other researchers to replicate the study by following your "recipe".
2. The design is the overall structure (variables, how they were manipulated or measured, operational definitions); the procedure is how the study was carried out, often told as what the participants did.
3. When there are complicated materials (several questionnaires, vignettes, perceptual stimuli); test-retest correlations, Cronbach's α or other reliability and validity statistics.
4. Save them and make them available to other researchers who request them; many journals encourage sharing them online.
5. Exclusions and the reasons for them; how multiple responses were combined into the main variables; whether the manipulation was successful.
6. Only step 3; the basic results should be clear in words, even to a reader who skips the numbers.
7. Two or three that could have influenced the results: explain how, and suggest ways to deal with them.
8. It can sound overreaching or banal and weaken the article; it is often better to return to the problem in the opening and say how the research addressed it.

2. Book exercise and sorting task

1. (example answer) Typical signals: summary, "The present study found that ..."; theoretical implications, "These results are consistent with ... theory"; practical implications, "These findings suggest that universities ..."; limitations, *may*, *might have*, *perhaps*; future research, "Future studies should ...". Not every discussion has all five.
2. (a) Participants; (b) nowhere (irrelevant); (c) Participants; (d) Materials; (e) Design; (f) Procedure; (g) nowhere (irrelevant); (h) Materials.

3. Language exercise

1. Participants will give their informed consent, read a set of instructions, complete a block of four practice trials, complete a block of 20 test trials, complete two questionnaires, and be debriefed and excused. (Repeating *will* before *be debriefed* is also correct.)
2. (a) might have produced (or could have produced); (b) may have responded (or might have responded).
3. c, e, b, d, a: remind (c), answer in words (e), statistics (b), qualify (d), summarize (a). (Checked: with 10 students in each condition, these means and standard deviations give $t(18) = 3.95$, $p = .0009$, reported as $p < .001$; $d = 1.86$ as in Lesson 28.)

My notes